UTM UNIVERSITI TEKNOLOGI MALAYSIA — Sekolah Pendidikan Profesional dan Pendidikan Berterusan (UTMSPACE)	Tutorial 2 CHEMISTRY II FSPC 0034	

1. Given a reaction: NH_4^+ (aq) + NO_2^- (aq) → N_2 (g) + $2H_2O$ (l)

Experiment	$[NH_4^+]$, mol L^{-1}	$[NO_2^-]$, mol L^{-1}	Rate, mol L^{-1} s^{-1}
1	0.35	0.35	8.0×10^{-4}
2	0.70	0.35	3.2×10^{-3}
3	0.70	0.70	6.4×10^{-3}

Based on the reaction and table above:

i. Determine the order of the reaction for each reactant and the overall order of reaction
ii. Calculate the rate constant, k

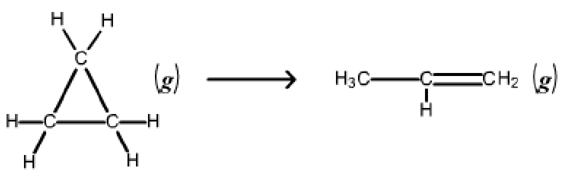
2. Given a reaction: 2 H_2 (g) + 2 NO (g) → N_2 (g) + $2H_2O$ (l)

Experiment	$[H_2]$ (M)	[NO] (M)	Initial Rate (M/s)
1	2.0×10^{-3}	5.0×10^{-3}	1.3×10^{-5}
2	2.0×10^{-3}	10.0×10^{-3}	5.0×10^{-5}
3	4.0×10^{-3}	10.0×10^{-3}	10.0×10^{-5}

Based on the reaction and table above:

i. Determine the rate law expression
ii. Calculate the rate constant, k

3. Cyclopropane (C_3H_6) is the smallest cyclic hydrocarbon. It is thermally unstable and rearranges to propene at 1000°C via the following first order reaction.



If the rate constant is 8.2 s^{-1}, calculate the half-life of the reaction.

4. Hydrogenation of cyclohexene form a cyclohexane in the presence of Pd/C as catalyzer. Cyclohexane is used in various solvent applications. It is volatile with a petroleum-like odor. The reaction of hydrogenation is given below at 75°C via the following first order reaction.

1) $\frac{Rate\ 2}{Rate\ 1} = \frac{k[NH_4^+]^x[NO_2^-]^y}{k[NH_4^+]^x[NO_2^-]^y}$

$\frac{3.2\times10^{-3}}{8.0\times10^{-4}} = \frac{0.70^x}{0.35^x}$

$4 = 2^x$

$x = 2$

$\frac{Rate\ 3}{Rate\ 2} = \frac{k[NH_4^+]^x[NO_2^-]^y}{k[NH_4^+]^x[NO_2^-]^y}$

$\frac{6.4\times10^{-3}}{3.2\times10^{-3}} = \frac{k(0.7)^2(0.7)^y}{k(0.7)^2(0.35)^y}$

$2 = \left(\frac{0.7}{0.35}\right)^y$

$y = 1$

overall order of reaction = 2 + 1 = 3

$8.0\times10^{-4} = k(0.35)^2(0.35)^1$

$k = 0.0186$

2) rate law expression = $k[H_2]^1[NO]^2$

$\frac{Rate\ 2}{Rate\ 1} = 2^y$

$\frac{5.0\times10^{-3}}{1.3\times10^{-3}} = 2^y$

$2^y = 3.85$

$y = \log_2 3.85$

$y = 1.94$

≈ 2

$\frac{Rate\ 3}{Rate\ 2} = \frac{(4.0\times10^{-3})^x}{(2.0\times10^{-3})^x}$

$2 = 2^x$

$x = 1$

$10\times10^{-5} = k(10\times10^{-3})(4\times10^{-3})$

$10 = 100k(4\times10^{-3})$

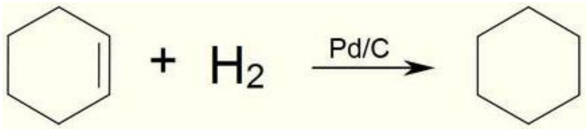
$4\times10^{-2}k = 1$

$k = 25$

3) $t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{8.2} = 0.0845\ s$

4) $t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{5.36\times10^{-4}} = 1292.91\ s$

$= 21.55\ min$



If the rate constant is $5.36 \times 10^{-4}\ s^{-1}$, calculate the half-life of the reaction in minute.

5. A flask is charged with 0.100 M of **A** and allowed to decompose to form **B** according to the hypothetiocal gas-phase reaction **A (g)** → **B (g)**. The following data are collected:

Time (s)	0	40	80	120	160
Molarity of A (M)	0.100	0.091	0.083	0.077	0.071

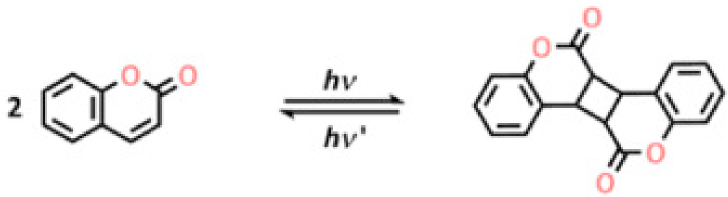
i. Calculate the average rate of disappearance of A from 40 s to 120 s time.
ii. Determine the order of this reaction.
iii. Determine the rate of reaction at 60 s.

6. Given a two steps reaction equations:

$$H_2O_2(aq) + I^-(aq) \rightarrow H_2O(l) + IO^-(aq) \qquad \text{SLOW}$$
$$IO^-(aq) + H_2O_2(aq) \rightarrow H_2O(l) + O_2(g) + I^-(aq) \qquad \text{FAST}$$

i. State the intermediate in the reaction
ii. Determine the rate law of the reaction

7. Photodimerization is a chemical process when one photoexcited molecule reacts with one unexcited molecule of the same species. One example of this process is the reaction of coumarin as shown below.



i. This reaction is reported as a second order reaction. Write a rate law equation for the forward reaction.
ii. Based on your answer in (i), calculate the initial concentration of coumarin, coumarin]$_0$ when k is 0.137 L/mol.s and initial rate is found to be 3.4×10^{-4} mol/L.s.

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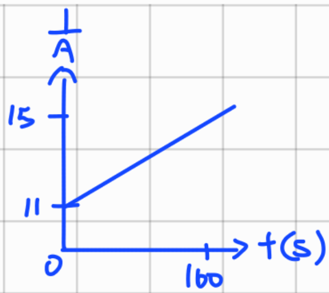
5) (i) $rate = -\frac{d[A]}{dt} \times \frac{1}{a}$

$= -\frac{0.077-0.091}{120-40} \times \frac{1}{1}$

$= 1.75 \times 10^{-4}$

(ii)



∴ second order

(iii) k = slope of the graph

$slope = \frac{\Delta y}{\Delta x} = \frac{\Delta(\frac{1}{A})}{\Delta t} = \frac{\frac{1}{0.071} - \frac{1}{0.1}}{160-0} = 0.025$

$\frac{1}{[A_T]} - \frac{1}{[A_0]} = kt$

$\frac{1}{[A_T]} - \frac{1}{0.1} = 0.025(60)$

$[A_T] = 0.087 M$

Rate of Reaction $= k[A]^2$

$= 0.025(0.087)^2$

$= 1.89 \times 10^{-4} Ms^{-1}$

6) Rate law: $rate = k[H_2O_2][I^-]$

order of reaction is second order

reaction intermediate is IO^- ion

rate law of reaction = rate of the slowest step

7) Rate law: $rate = k[C_4H_6O_2]^2$

$3.4 \times 10^{-4} = 0.137 x^2$

$x = 0.05 M$

8) (i) $Rate = k[CH_3Br]^1[OH^-]^1$

$0.256 = k(5 \times 10^{-3})(0.12)$

$k = 426.67 s^{-1}$

(ii) unit of rate constant : s^{-1}

9) (i) the reaction is second order because $\frac{1}{[NO_2]}$ versus time gives a linear graph with positive slope which represent the characteristic of a second order reaction

(ii) $t_{\frac{1}{2}} = \frac{1}{k[NO_2]_0}$

$= \frac{1}{10.14(0.1)}$

$= 0.986 s$

$\frac{1}{[NO_2]_0} - \frac{1}{[NO_2]} = kt$

$\frac{1}{0.0047} - \frac{1}{0.1} = k(20-0)$

$k = 10.14 M^{-1}s^{-1}$

(iii) yes, because the gradient, k of the graph decreased constantly

8. Consider the following reaction:

$$CH_3Br\ (aq) + OH^-\ (aq) \rightarrow CH_3OH(aq) + Br^-(aq)$$

The rate law for this reaction is first order in CH_3Br and first order in OH^-. When $[CH_3Br]$ is 5.0 X 10^{-3} M and $[OH^-]$ is 0.120 M, the reaction rate at 303 K is 0.256 M/s.

i. Calculate the rate constant

ii. Determine the unit of the rate constant

9. The gas phase decomposition of NO_2, $2\ NO_2(g) \rightarrow 2\ NO\ (g) + O_2(g)$, is studied at 383 °C, giving the following data:

Time (s)	**$[NO_2]$ (M)**	ln	$\frac{1}{x}$
0.0	0.100	-2.3	10
5.0	0.017	-4.1	58
10.0	0.0090	-4.7	111
15.0	0.0062	-5.1	161
20.0	0.0047	-5.3	212

i. Determine the order of the reaction by using the mathematical method.

ii. Calculate the half life of the reaction.

iii. Do you expect the second half-life of the reaction to have the same value as the first one? Briefly explain your reasoning.